

## Equilibrium Index

### Problem

Given an array of  $N$  integers, find the smallest index  $i$  such that the sum of all elements to its left is equal to the sum of all elements to its right. This index is called the Equilibrium Index.

If no such index exists, print -1.

### Example 1

Input:  $\text{arr} = \{-7, 1, 5, 2, -4, 3, 0\}$

**Output: 3**

Explanation: At index 3 (value 2): left sum =  $-7+1+5 = -1$ , right sum =  $-4+3+0 = -1$ . Equal, so index 3 is the answer.

### Example 2

Input:  $\text{arr} = \{1, 2, 3\}$

**Output: -1**

Explanation: No index has equal left and right sums, so the answer is -1.

**Hint:** Compute the total sum first, then move left to right while maintaining a running left-sum; at each index,  $\text{right-sum} = \text{total} - \text{leftSum} - \text{arr}[i]$ .